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Impact of the high-frequency cutoff of bandpass filtering on ECG quality and clinical interpretation: A comparison between 40 Hz and 150 Hz cutoff in a surgical preoperative adult outpatient population☆

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Abstract

Background: In 1990 the American Heart Association (AHA) established a standard 0.05 to 150 Hz bandwidth for the routine recording of 12-lead electrocardiograms (ECGs). However, subsequent studies have indicated a very high prevalence of deviations from the recommended cutoffs.
Objective: This prospective observational study investigates the impact of 40 Hz compared to 150 Hz high-frequency cutoffs on ECG quality and clinical interpretation in a single-center surgical outpatient population.
Methods: 1582 consecutive adult patients underwent two standard 12-lead ECG tracings using different high-frequency cutoffs (40 Hz and 150 Hz). Two blinded cardiologists randomly reviewed and interpreted the recordings according to pre-defined parameters (PR and ST segment, Q and T wave abnormalities). An arbitrary score, ranging from 1 to 3, was established to evaluate the perceived quality of the recordings and the non-interpretable ECGs were noted. The tracings were then matched to compare interpretations between 40 and 150 Hz filters.
Results: A 40 Hz high-frequency cutoff resulted in an increased rate of optimal quality ECGs compared to the 150 Hz cutoff (93.4% vs 54.6%; $p < 0.001$) and a lower rate of non-interpretable traces (0.25% vs 4.80%; $p < 0.001$). Analyzing the morphologic parameters, no significant differences between the filter settings were found, except for a higher incidence of the J-point elevation in the 40 Hz high-frequency cutoff ($p = 0.007$) and a higher incidence of left ventricular hypertrophy in the 150 Hz high-frequency cutoff (7.4% vs 5.4%, $p < 0.001$). The latter was noted only in ECGs with borderline QRS amplitudes (between 3.3 and 3.7 mV; $p < 0.001$).
Conclusion: Despite current recommendations, the large deviation from standard high-frequency cutoff in clinical practice does not seem to significantly affect ECG clinical interpretation and a 40 Hz high-frequency cutoff of the band-pass filtering may be acceptable in a low risk population, allowing for a better quality of tracings.
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Keywords: ECG; High-frequency ECG filtering; Left ventricular hypertrophy; Digital signal processing

Introduction

Electrocardiography is a commonly used, noninvasive procedure for recording electrical changes in the heart, detecting arrhythmias and conduction disturbances; it is a safe, simple, reproducible procedure with minimal relative costs. In the preoperative evaluation, patients are often screened only on the basis of an ECG, that is frequently interpreted without a clinical assessment of the patients, hence missing out on personal medical history, symptoms and physical examination.

The quality of recorded tracings may be altered by a number of technical factors: patient-related (i.e., muscle tremors and movement), operator-dependent (i.e., incorrect

Abbreviations: AHA, American Heart Association; LVH2, left ventricular hypertrophy.

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position of electrodes or inadequate skin contact) and those related to the equipment (i.e., standard settings including processing of the signal or current leakage).

The American Heart Association (AHA) established a standard 0.05 to 150 Hz bandwidth for the routine recording of 12-lead ECGs, where "bandwidth" refers to the low- to high-frequency content of the recorded signal [1,2]. Greater bandwidth may be required for accurate determination of amplitudes in infants. The bandwidth of an electrocardiograph influences the fidelity of electrocardiographic waveforms, including the amplitudes used for the diagnosis of ventricular hypertrophy, the accuracy of the magnitudes of ST-segment modifications, Q-wave measurements [3–5], identification of such QRS aspects as the presence of notching. Previous studies have indicated, in everyday clinical practice, a very high prevalence of deviations from the recommended filter settings (up to 75%) mainly because of improved tracing appearance; however, those arbitrary settings can potentially modify the clinical interpretation [6].

The aim of this study is to investigate whether the commonly used reduced high-frequency cutoff (40 Hz) can have potential clinical consequences on ECG interpretation compared to the standard cutoff (150 Hz), among a preoperative ECG series in a single-center population.

Methods

This prospective observational study enrolled consecutive adult outpatients undergoing routine preoperative ECG in our institution between October 2014 and April 2015. The study was approved by the Institutional Committee on Human Research at the authors' institution. All records were collected with the same electrocardiograph (ELI 250C, Mortara instruments Inc., Milwaukee, WI, USA). Nurses were trained to print two standard 12-lead ECG tracings for each patient: one with a high-frequency cutoff of 40 Hz and another with a high-frequency cutoff of 150 Hz. The low-frequency cutoff was set at 0.05 Hz. Baseline characteristics of patients were collected.

Two blinded experienced cardiologists reviewed and interpreted all the ECG tracings according to qualitative and quantitative parameters. Of note, the traces were reviewed in a casual order to reduce potential biases and disagreements were resolved by consensus (e.g., the QRS amplitude was re-measured in case of disagreement between the reviewers).

The following parameters were considered and compared: PR segment and ST-T wave abnormalities; Q-wave amplitude (>0.1 mV or suggestive of myocardial necrosis) [7]; J-point elevation; and visible delta-waves, epsilon-waves and pacemaker spikes. The resulting QRS amplitude was manually measured adding the deepest S wave in V1–V2 and the highest R wave in V5–V6. Left ventricular hypertrophy (LVH) was assessed using the Sokolow–Lyon, i.e., resulting QRS amplitude ≥3.5 mV). J point elevation is defined as an elevation ≥0.1 mV with either a notch inscribed on the downslope of a prominent R wave or a slur (a smooth transition from QRS to ST segment) ≥0.1 mV in two contiguous leads [8]. The delta-wave was defined as a slurred upstroke to the QRS complex and the epsilon-wave as a small positive deflection at the end of the QRS complex visible in V1.

Table 1
Baseline characteristics of the study population.

Characteristics	Study population (N = 1582)
Age (y)	57.8 ± 16.9
Male sex	660 (41.7%)
Sinus rhythm	1542 (97.5%)
Atrial fibrillation	35 (2.2%)
Supraventricular extrasystoles	55 (3.5%)
Ventricular extrasystoles	35 (2.2%)
Heart rate (bpm)	68.2 ± 11.5
PR interval (ms)	161.8 ± 29.6
QRS complex (ms)	98.6 ± 18.8
QT interval (ms)	393.2 ± 31.6
QT_c interval (ms)	407.7 ± 24.0
Right bundle branch block	117 (7.4%)
Left bundle branch block	39 (2.5%)

The ECG data were derived from the standard 150 Hz high-frequency cutoff tracing. The PR, QRS and QT interval were automatically measured (from the standard 150 Hz high-frequency cutoff tracing).

In order to assess the perceived quality of tracings, the physicians were asked to evaluate the traces with an arbitrary score, ranging from 1 to 3, where 1 indicated a poor quality trace, 2 indicated an average and acceptable quality and 3 indicated an optimal quality for clinical interpretation. Poor quality traces (Score 1) were also discerned as interpretable or non-interpretable (i.e., presence of artifacts and/or noise not permitting a clear detection of waves and intervals), thus requiring the repetition of the ECG.

Statistical analysis

Assumptions of normality were checked and met for continuous variables (expressed as mean ± standard deviation), which were then compared with paired Student t-test. Binary variables were expressed as absolute frequencies followed by percentage (%) and were compared with McNemar test or chi-square test, as appropriate. Quality of tracings was compared using Wilcoxon signed-rank test for paired data. A p value less than 0.01 was considered statistically significant. Statistical analysis was performed using SPSS 19.0 (SPSS, Inc., Chicago, Illinois).

Results

A total of 1582 patients were included in the study. Baseline demographic and ECG data are summarized in Table 1. The mean age of the study participants was 57.8 ± 16.9 years (range 17–94) and 41.7% were male. The majority of traces showed sinus rhythm (97.5%), 7.2% of patients presented with first-degree atrio-ventricular block, 7.4% had a right bundle branch block and 2.5% had a left bundle branch block.

Analyzing the morphologic parameters, we found QRS amplitude to be higher in the 150 Hz group (2.31 ± 0.76 vs 2.11 ± 0.74 mV; $p < 0.001$). A higher incidence of J-point elevation and a lower incidence of LVH were found in the 40-Hz compared to the 150-Hz high-frequency cutoff

Table 2
ECG findings according to filter setting (percentage in brackets).

	40 Hz high-frequency cutoff	150 Hz high-frequency cutoff	*p* Value
PR abnormalities	5 (0.3)	7 (0.4)	0.500
ST abnormalities	481 (30.4)	476 (30.1)	0.267
T-wave abnormalities	446 (28.2)	448 (28.3)	0.791
Q-wave >0.1 mV	183 (11.6)	144 (9.1)	0.804
Significant Q-wave[a]	79 (5.0)	69 (4.4)	1.000
J-point elevation	166 (10.5)	152 (9.6)	0.007
ε-Wave	–	–	–
δ-Wave	–	–	–
Pacemaker spikes	16 (1.0)	15 (0.9)	1.000
QRS amplitude (mV)	2.11 ± 0.74	2.31 ± 0.76	<0.001
Left Ventricular hypertrophy	86 (5.4)	117 (7.4)	<0.001

[a] Suggestive of myocardial necrosis.

(respectively, 10.5% vs 9.6%, $p = 0.007$ and 5.4% vs 7.4%, $p < 0.001$); no other significant differences between the filter settings were found in the ECG interpretation (Table 2). Thirty-one patients (2.0%) were diagnosed with LVH using the 150 Hz filter, but did not meet Sokolow–Lyon criterion in the 40 Hz recordings; conversely, the opposite situation was observed in two patients. In 84 patients, the diagnosis of LVH was concordant between the two modalities. The entire population was then divided on the basis of resulting QRS amplitudes: <3.2 mV, 3.3–3.7 mV and >3.8 mV. Among 105 patients with resulting QRS amplitude between 3.3 and 3.7 mV, a significant difference in the incidence of LVH (amplitude ≥3.5 mV) was found, compared to the 40-Hz high-frequency cutoff, in 31 patients (2.0% of the entire study population; $p < 0.001$). Instead, in patients with a resulting QRS amplitude <3.2 mV or >3.8 mV, no differences were found, even in the incidence of LVH, compared to the reduced high-frequency cutoff.

The 40-Hz high-frequency cutoff resulted in an increased rate of optimal quality ECGs compared to the 150-Hz cutoff (93.4% vs 54.6%; $p < 0.001$) (Figs. 1 and 2). Regarding the traces judged of poor quality (score 1), 4 out of 10 (40%) in the 40-Hz group and 76 out of 159 (47.8%) in the 150 Hz group were classified as non-interpretable. Rate of non-interpretable traces was 0.25% in the 40-Hz group and 4.8% in the 150-Hz group ($p < 0.001$).

Discussion

The main finding of this study is that the elimination of the frequencies between 40 Hz and 150 Hz does not substantially affect ECG interpretation in the pre-operative setting, with the exception of QRS amplitude. To the best of our knowledge, this is the first study assessing the use of a reduced high-frequency cutoff and its clinical implications in a large cohort of patients, in routine clinical practice.

Although most of the power in the QRS waveform is below 40 Hz, old studies have demonstrated that high-frequency cutoffs of 150 Hz or even higher are required to accurately represent amplitudes and durations of depolarization [9,10]. In brief, a filter is designed to attenuate or remove some frequencies from an input data; however, it should only remove the noise without changing the desired signal [11]. Unfortunately, noise and desired signals often overlap in the frequency domain. As a result, when a filter attenuates the frequency components of noise, the overlapping desired signal can also be impacted.

Although a reduced high-frequency cutoff can minimize motion and tremor artifacts, previous studies have clearly showed that it could attenuate the QRS complex, Q-wave measurements and smooth small notches [1,7–12]. The right compromise between quality of tracings appearance and quality of ECG data should be achieved. Perceived quality and appearance of ECG tracings are in fact fundamental to ensure an accurate clinical interpretation of waveforms.

Our study is the first to demonstrate that a better perceived quality of 40 Hz filter ECG traces renders recordings with high frequency noise clinically acceptable, with a lower rate of non-interpretable tracings compared with 150 Hz filtering.

Nakagawa et al. [13] showed on 665 patients that the ECG filter setting significantly affects the J-wave morphology; in particular, its amplitude was significantly lower with 25- and 35-Hz filters than with 75-, 100-, and 150-Hz filters. In contrast, we found a higher incidence of J-point elevation ≥0.1 mV in the 40 Hz ECGs. Even though there are previous studies in literature showing that reducing the ECG bandwidth can reduce the ability to detect QRS notches [14], according to the authors' opinion, the improved quality of the ECG tracings allows for a better evaluation of the morphology of the terminal part of QRS, not only leading to a lower rate of mis-diagnosed J waves, but also improving its identification. The clinical significance of this result is of great importance, e.g., in the setting of an acute coronary syndrome (ST elevation ≥1 mm in two contiguous derivations other than V2–V3) [7]. A better definition of QRS notching may be of great importance in the diagnosis of many cardiac conditions. For example, the presence of a notch in the ascending part of the S wave in V3 or V4 (Cabrera's sign) has a high specificity for identification of a previous myocardial infarction in patients with complete left bundle brunch block. The presence of a notched QRS may also have a role in the risk stratification in the setting of early repolarization [15]. Another issue is represented by the identification of fragmented QRS, which has been associated with myocardial scar, for example, in patients with coronary artery disease and had a prognostic role in this setting [16]. Our study was not designed to identify a fragmented QRS and further studies are warranted to evaluate if the use of 40 Hz cutoff be an adequate approach to this aim.

The QRS amplitude significantly differs between the two cutoffs, resulting in a higher rate of LVH detected. However, the significant difference in LVH criteria comprises only the individuals with borderline QRS amplitudes (between 3.3 and 3.7 mV), a minor part of the population evaluated. The clinical impact of this result is minimal when compared to the whole study cohort, as it has been already demonstrated that LVH diagnosis should not be solely based on a pure measurement of QRS amplitude and that ECG has a low sensitivity and an unacceptable low positive predictive value when compared to the echocardiographic measurement [17].

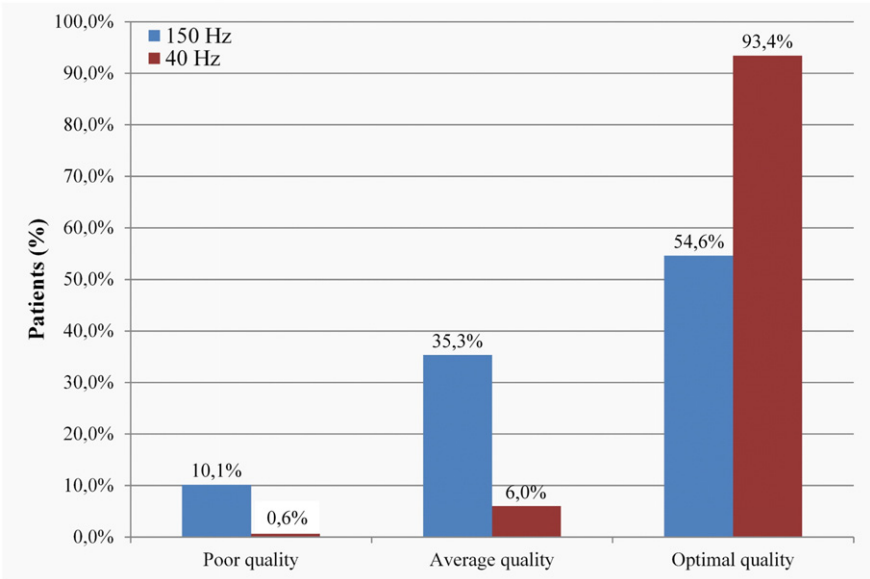

Fig. 1. Quality of the ECG tracing according to filtering.

Conventional ECG criteria for LVH are based on recordings with a 150 Hz high frequency cutoff and it is unclear how they could be affected by reducing frequency to 40 Hz. It should be kept in mind, however, that none of the available ECG criteria for LVH have a sufficient diagnostic accuracy when compared with the gold standard (i.e., echocardiography) [18].

Moreover, Luo et al. [19] highlighted that a reduced high-frequency cutoff could filter out the spike presentations from ECG printouts. Conversely, this study failed in showing any differences in pacemaker spikes evidence between the two filters, although the prevalence of the event was too low to draw definitive conclusions (approximately 1% of the study population).

Based on studies published in the 80s, current recommendations [1,2] suggest a high-frequency cutoff of 150 Hz for ECG recording. However, considering the technological evolution of the latest decades, ECG recording with modern machines, may potentially overcome this limitation. In fact, unconventional filtering is nowadays frequently used to

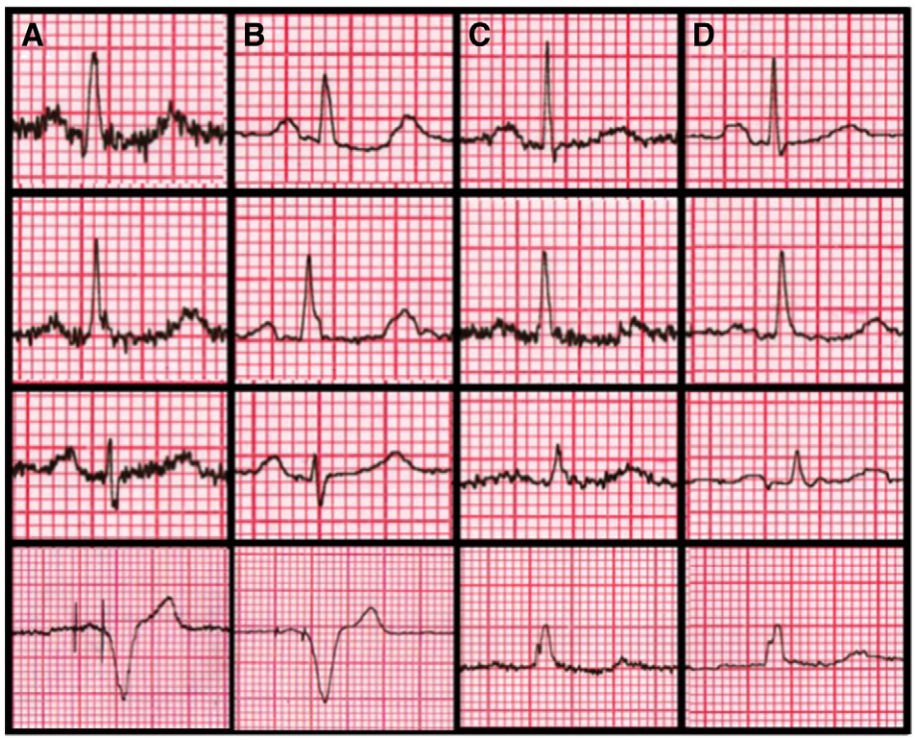

Fig. 2. Some examples of ECG traces performed at 150 Hz high-frequency cutoff (columns A and C) and at 40 Hz high-frequency cutoff (columns B and D). The 2nd row of column A shows a distorted QRS where it is difficult to evaluate the terminal part of QRS (slur or artifacts?), while the same row of column B allows for a better discrimination of a slur in the QRS, with a J-point elevation of 0.2 mV. The ECG traces at 150 Hz high-frequency cutoff of bandpass filtering (with the exception of the one with pacemaker) were judged of poor quality (score 1), but still interpretable by the readers.

obtain cleaner traces [6] and, according to our data, no significant alteration in tracing interpretation should be expected when using 40 Hz filtering in a low risk population. We believe that, based on these results, a revision of the recommendations for standardization and interpretation of ECG may be warranted.

Limitations

The main limitation of this study is the low prevalence of some ECG findings in the general as well as in the study population (i.e., delta-wave, epsilon-wave, pacemaker spikes), which does not allow drawing firm conclusions on the topic.

LVH diagnosis was only based on ECG recording, not confirmed with an echocardiographic evaluation because the majority of the patients did not undergo a second level cardiologic evaluation.

Finally, all ECGs tracings were recorded by the same ECG machine (ELI 250C, Mortara Instruments Inc.) and in the same environment (single-center study), which does not allow for a direct comparison with other electrocardiograph brands and it does not take account of potential differences in electrical and magnetic fields interferences.

LVH diagnosis has been made assessing the Sokolow–Lyon criterion solely; no other criteria have been evaluated or analyzed.

Conclusion

Despite current recommendations, the large deviation from the standard high-frequency cutoff in clinical practice does not seem to significantly affect ECG interpretation and may be acceptable in a low-risk surgical preoperative adult outpatient population, possibly allowing for a better perceived quality of tracings. The adoption of different high frequency filtering could be useful with particular clinical suspicions.

Funding and conflict of interest

None.

Clinical perspectives

This study demonstrates that, with newer ECG machines, the standardization of a 40 Hz cutoff filtering improves the quality of a 12-lead ECG without significant impact on the diagnostic potential of this invaluable cardiologic tool. Future researches in ECG filtering and analysis methods will clarify the optimal setting of the machines for a safe and comprehensive ECG evaluation.

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